Coeficientes de Clebsch–Gordan

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Link al repositorio:
`https://github.com/s4ul19/Clebsch--Gordan-Coefficients`

1. Qué son los coeficientes de Clebsch–Gordan

Al sumar dos momentos angulares, $\mathbf{J}_1$ y $\mathbf{J}_2$, el espacio de estados es el producto tensorial $\mathcal{H}_{j_1} \otimes \mathcal{H}_{j_2}$. Una base natural es la *base producto*, formada por $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$, que distingue las proyecciones de ambos momentos por separado. Otra es la *base acoplada*, $|J, M\rangle$, que diagonaliza el momento angular total $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$:

$$J^2 \,|J, M\rangle = J(J+1)\,|J, M\rangle\,, \qquad J_z\,|J, M\rangle = M\,|J, M\rangle\,. \tag{1}$$

Aquí se adopta $\hbar = 1$.

Los coeficientes de Clebsch–Gordan son las entradas de la matriz de cambio de base entre la base producto y la base acoplada. A cada $|J, M\rangle$ le corresponde una combinación lineal de vectores de la base producto; las amplitudes de esta superposición son los coeficientes de Clebsch–Gordan:

$$|J, M\rangle = \sum_{m_1, m_2} C^{JM}_{j_1 m_1, j_2 m_2}\; |j_1, m_1\rangle \otimes |j_2, m_2\rangle\,, \qquad C^{JM}_{j_1 m_1, j_2 m_2} = \big(\langle j_1, m_1| \otimes \langle j_2, m_2|\big)\,|J, M\rangle\,. \tag{2}$$

Los valores permitidos del momento total son $J = |j_1 - j_2|, |j_1 - j_2| + 1, \ldots, j_1 + j_2$.

2. Un método para determinar los coeficientes

El cálculo consiste en construir las matrices del momento angular total en la base producto y encontrar sus estados propios. Se adopta el orden $m_1 = j_1, j_1 - 1, \ldots, -j_1$ y $m_2 = j_2, j_2 - 1, \ldots, -j_2$.

2.1. Construcción de los operadores

Para cada momento angular j, las matrices se determinan a partir de

$$J_z\,|j, m\rangle = m\,|j, m\rangle\,, \qquad J^2\,|j, m\rangle = j(j+1)\,|j, m\rangle\,, \tag{3}$$

$$J_\pm\,|j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)}\,|j, m \pm 1\rangle\,. \tag{4}$$

Los operadores J_+ y J_- elevan y disminuyen la proyección m en una unidad. Para extenderlos al espacio producto se emplean identidades sobre el otro subsistema; por ejemplo,

$$J_{1z} = J_z^{(j_1)} \otimes \mathbb{I}_{2j_2+1}, \qquad J_{2z} = \mathbb{I}_{2j_1+1} \otimes J_z^{(j_2)}. \tag{5}$$

De la suma $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$ se obtiene

$$J_z = J_{1z} + J_{2z}, \qquad J_\pm = J_{1\pm} + J_{2\pm}, \tag{6}$$

$$J^2 = J_1^2 + J_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+}. \tag{7}$$

2.2. Identificación de los coeficientes

Las componentes de cada vector normalizado $|J, M\rangle$ en la base producto son los coeficientes de la ecuación (2). En particular,

$$C^{JM}_{j_1 m_1, j_2 m_2} = 0 \quad \text{si } m_1 + m_2 \neq M, \qquad \sum_{m_1, m_2} \left|C^{JM}_{j_1 m_1, j_2 m_2}\right|^2 = 1. \tag{8}$$

Por ejemplo, para $j_1 = 2$ y $j_2 = \frac{1}{2}$:

$$\left|\tfrac{5}{2}, \tfrac{3}{2}\right\rangle = \frac{1}{\sqrt{5}} \left|2, 2\right\rangle \otimes \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle + \frac{2}{\sqrt{5}} \left|2, 1\right\rangle \otimes \left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle, \tag{9}$$

$$\left|\tfrac{3}{2}, \tfrac{3}{2}\right\rangle = \frac{2}{\sqrt{5}} \left|2, 2\right\rangle \otimes \left|\tfrac{1}{2}, -\tfrac{1}{2}\right\rangle - \frac{1}{\sqrt{5}} \left|2, 1\right\rangle \otimes \left|\tfrac{1}{2}, \tfrac{1}{2}\right\rangle. \tag{10}$$

Los dos estados son ortogonales y están normalizados. Multiplicar un estado completo por -1 cambia los signos de sus coeficientes, pero representa el mismo estado físico.

3. Estados acoplados en la base producto

3.1. $j_1 = \frac{1}{2}$ y $j_2 = \frac{1}{2}$

$$|1,1\rangle = \left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{2}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|0,0\rangle = \frac{1}{\sqrt{2}}\left|\frac{1}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|1,-1\rangle = \left|\frac{1}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle$$

3.2. $j_1 = 1$ y $j_2 = \frac{1}{2}$

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = |1,1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}|1,1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}|1,0\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}|1,1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{1}{\sqrt{3}}|1,0\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}|1,0\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{3}}|1,-1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{3}}|1,0\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}|1,-1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{3}{2}\right\rangle = |1,-1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle$$

3.3. $j_1 = 1$ y $j_2 = 1$

$$|2,2\rangle = |1,1\rangle \otimes |1,1\rangle$$

$$|2,1\rangle = \frac{1}{\sqrt{2}}|1,1\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{2}}|1,0\rangle \otimes |1,1\rangle$$

$$|1,1\rangle = \frac{1}{\sqrt{2}}|1,1\rangle \otimes |1,0\rangle - \frac{1}{\sqrt{2}}|1,0\rangle \otimes |1,1\rangle$$

$$|2,0\rangle = \frac{1}{\sqrt{6}}|1,1\rangle \otimes |1,-1\rangle + \sqrt{\frac{2}{3}}|1,0\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{6}}|1,-1\rangle \otimes |1,1\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}|1,1\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{2}}|1,-1\rangle \otimes |1,1\rangle$$

$$|0,0\rangle = \frac{1}{\sqrt{3}}|1,1\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{3}}|1,0\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{3}}|1,-1\rangle \otimes |1,1\rangle$$

$$|2,-1\rangle = \frac{1}{\sqrt{2}}\,|1,0\rangle \otimes |1,-1\rangle + \frac{1}{\sqrt{2}}\,|1,-1\rangle \otimes |1,0\rangle$$

$$|1,-1\rangle = \frac{1}{\sqrt{2}}\,|1,0\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{2}}\,|1,-1\rangle \otimes |1,0\rangle$$

$$|2,-2\rangle = |1,-1\rangle \otimes |1,-1\rangle$$

3.4. $j_1 = 2$ y $j_2 = \frac{1}{2}$

$$\left|\frac{5}{2},\frac{5}{2}\right\rangle = |2,2\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},\frac{3}{2}\right\rangle = \frac{1}{\sqrt{5}}\,|2,2\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{2}{\sqrt{5}}\,|2,1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \frac{2}{\sqrt{5}}\,|2,2\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{1}{\sqrt{5}}\,|2,1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\,|2,1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{3}{5}}\,|2,0\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{3}{5}}\,|2,1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{5}}\,|2,0\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{3}{5}}\,|2,0\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{5}}\,|2,-1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\,|2,0\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{3}{5}}\,|2,-1\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{3}{2}\right\rangle = \frac{2}{\sqrt{5}}\,|2,-1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}\,|2,-2\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{3}{2}\right\rangle = \frac{1}{\sqrt{5}}\,|2,-1\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{2}{\sqrt{5}}\,|2,-2\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{5}{2}\right\rangle = |2,-2\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle$$

3.5. $j_1 = \frac{3}{2}$ y $j_2 = \frac{1}{2}$

$$|2,2\rangle = \left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|2,1\rangle = \frac{1}{2}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{\sqrt{3}}{2}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|1,1\rangle = \frac{\sqrt{3}}{2}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{1}{2}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|2,0\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|2,-1\rangle = \frac{\sqrt{3}}{2}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle + \frac{1}{2}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|1,-1\rangle = \frac{1}{2}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle - \frac{\sqrt{3}}{2}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},\frac{1}{2}\right\rangle$$

$$|2,-2\rangle = \left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{1}{2},-\frac{1}{2}\right\rangle$$

3.6. $j_1 = 2$ y $j_2 = 1$

$$|3,3\rangle = |2,2\rangle \otimes |1,1\rangle$$

$$|3,2\rangle = \frac{1}{\sqrt{3}}|2,2\rangle \otimes |1,0\rangle + \sqrt{\frac{2}{3}}|2,1\rangle \otimes |1,1\rangle$$

$$|2,2\rangle = \sqrt{\frac{2}{3}}|2,2\rangle \otimes |1,0\rangle - \frac{1}{\sqrt{3}}|2,1\rangle \otimes |1,1\rangle$$

$$|3,1\rangle = \frac{1}{\sqrt{15}}|2,2\rangle \otimes |1,-1\rangle + 2\sqrt{\frac{2}{15}}|2,1\rangle \otimes |1,0\rangle + \sqrt{\frac{2}{5}}|2,0\rangle \otimes |1,1\rangle$$

$$|2,1\rangle = \frac{1}{\sqrt{3}}|2,2\rangle \otimes |1,-1\rangle + \frac{1}{\sqrt{6}}|2,1\rangle \otimes |1,0\rangle - \frac{1}{\sqrt{2}}|2,0\rangle \otimes |1,1\rangle$$

$$|1,1\rangle = \sqrt{\frac{3}{5}}|2,2\rangle \otimes |1,-1\rangle - \sqrt{\frac{3}{10}}|2,1\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{10}}|2,0\rangle \otimes |1,1\rangle$$

$$|3,0\rangle = \frac{1}{\sqrt{5}}|2,1\rangle \otimes |1,-1\rangle + \sqrt{\frac{3}{5}}|2,0\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{5}}|2,-1\rangle \otimes |1,1\rangle$$

$$|2,0\rangle = \frac{1}{\sqrt{2}}|2,1\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{2}}|2,-1\rangle \otimes |1,1\rangle$$

$$|1,0\rangle = \sqrt{\frac{3}{10}}|2,1\rangle \otimes |1,-1\rangle - \sqrt{\frac{2}{5}}|2,0\rangle \otimes |1,0\rangle + \sqrt{\frac{3}{10}}|2,-1\rangle \otimes |1,1\rangle$$

$$|3,-1\rangle = \sqrt{\frac{2}{5}}|2,0\rangle \otimes |1,-1\rangle + 2\sqrt{\frac{2}{15}}|2,-1\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{15}}|2,-2\rangle \otimes |1,1\rangle$$

$$|2,-1\rangle = \frac{1}{\sqrt{2}}|2,0\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{6}}|2,-1\rangle \otimes |1,0\rangle - \frac{1}{\sqrt{3}}|2,-2\rangle \otimes |1,1\rangle$$

$$|1,-1\rangle = \frac{1}{\sqrt{10}}|2,0\rangle \otimes |1,-1\rangle - \sqrt{\frac{3}{10}}|2,-1\rangle \otimes |1,0\rangle + \sqrt{\frac{3}{5}}|2,-2\rangle \otimes |1,1\rangle$$

$$|3,-2\rangle = \sqrt{\frac{2}{3}}|2,-1\rangle \otimes |1,-1\rangle + \frac{1}{\sqrt{3}}|2,-2\rangle \otimes |1,0\rangle$$

$$|2,-2\rangle = \frac{1}{\sqrt{3}}|2,-1\rangle \otimes |1,-1\rangle - \sqrt{\frac{2}{3}}|2,-2\rangle \otimes |1,0\rangle$$

$$|3,-3\rangle = |2,-2\rangle \otimes |1,-1\rangle$$

3.7. $j_1 = \frac{3}{2}$ y $j_2 = 1$

$$\left|\frac{5}{2},\frac{5}{2}\right\rangle = \left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{5}{2},\frac{3}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,0\rangle + \sqrt{\frac{3}{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \sqrt{\frac{3}{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,0\rangle - \sqrt{\frac{2}{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{5}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{10}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,-1\rangle + \sqrt{\frac{3}{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle + \sqrt{\frac{3}{10}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,-1\rangle + \frac{1}{\sqrt{15}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle - 2\sqrt{\frac{2}{15}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{3}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{6}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{5}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{3}{10}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,-1\rangle + \sqrt{\frac{3}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{10}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = 2\sqrt{\frac{2}{15}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{15}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,0\rangle - \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{6}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes |1,-1\rangle - \frac{1}{\sqrt{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,0\rangle + \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,1\rangle$$

$$\left|\frac{5}{2},-\frac{3}{2}\right\rangle = \sqrt{\frac{3}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,-1\rangle + \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,0\rangle$$

$$\left|\frac{3}{2},-\frac{3}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes |1,-1\rangle - \sqrt{\frac{3}{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,0\rangle$$

$$\left|\frac{5}{2},-\frac{5}{2}\right\rangle = \left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes |1,-1\rangle$$

3.8. $j_1 = 2$ y $j_2 = \frac{3}{2}$

$$\left|\frac{7}{2},\frac{7}{2}\right\rangle = |2,2\rangle \otimes \left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{7}{2},\frac{5}{2}\right\rangle = \sqrt{\frac{3}{7}}\,|2,2\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{2}{\sqrt{7}}\,|2,1\rangle \otimes \left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{5}{2},\frac{5}{2}\right\rangle = \frac{2}{\sqrt{7}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \sqrt{\frac{3}{7}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{7}{2},\frac{3}{2}\right\rangle = \frac{1}{\sqrt{7}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{2}{\sqrt{7}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \sqrt{\frac{2}{7}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{5}{2},\frac{3}{2}\right\rangle = \frac{4}{\sqrt{35}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{35}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - 3\sqrt{\frac{2}{35}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{3}{2},\frac{3}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{5}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{7}{2},\frac{1}{2}\right\rangle = \frac{1}{\sqrt{35}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + 2\sqrt{\frac{3}{35}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + 3\sqrt{\frac{2}{35}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{2}{\sqrt{35}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{5}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{6}{35}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \sqrt{\frac{5}{14}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{3}{35}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - 3\sqrt{\frac{3}{70}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{3}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{5}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \sqrt{\frac{2}{5}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|2,2\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \sqrt{\frac{3}{10}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \frac{1}{\sqrt{10}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{7}{2},-\frac{1}{2}\right\rangle = \frac{2}{\sqrt{35}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + 3\sqrt{\frac{2}{35}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + 2\sqrt{\frac{3}{35}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{1}{\sqrt{35}}\left|2,-2\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{1}{2}\right\rangle = 3\sqrt{\frac{3}{70}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \sqrt{\frac{3}{35}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{5}{14}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \sqrt{\frac{6}{35}}\left|2,-2\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{5}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{5}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{5}}\left|2,-2\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = \frac{1}{\sqrt{10}}\left|2,1\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{5}}\left|2,0\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{3}{10}}\left|2,-1\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \sqrt{\frac{2}{5}}\left|2,-2\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$\left|\frac{7}{2},-\frac{3}{2}\right\rangle = \sqrt{\frac{2}{7}}\,|2,0\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \frac{2}{\sqrt{7}}\,|2,-1\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{7}}\,|2,-2\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{3}{2}\right\rangle = 3\sqrt{\frac{2}{35}}\,|2,0\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{35}}\,|2,-1\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \frac{4}{\sqrt{35}}\,|2,-2\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{3}{2},-\frac{3}{2}\right\rangle = \frac{1}{\sqrt{5}}\,|2,0\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \sqrt{\frac{2}{5}}\,|2,-1\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{5}}\,|2,-2\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$\left|\frac{7}{2},-\frac{5}{2}\right\rangle = \frac{2}{\sqrt{7}}\,|2,-1\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \sqrt{\frac{3}{7}}\,|2,-2\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

$$\left|\frac{5}{2},-\frac{5}{2}\right\rangle = \sqrt{\frac{3}{7}}\,|2,-1\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{2}{\sqrt{7}}\,|2,-2\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

$$\left|\frac{7}{2},-\frac{7}{2}\right\rangle = |2,-2\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle$$

3.9. $j_1 = \frac{3}{2}$ y $j_2 = \frac{3}{2}$

$$|3,3\rangle = \left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|3,2\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|2,2\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|3,1\rangle = \frac{1}{\sqrt{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{3}{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|2,1\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|1,1\rangle = \sqrt{\frac{3}{10}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \sqrt{\frac{2}{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \sqrt{\frac{3}{10}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|3,0\rangle = \frac{1}{2\sqrt{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \frac{3}{2\sqrt{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{3}{2\sqrt{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{1}{2\sqrt{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|2,0\rangle = \frac{1}{2}\left|\frac{3}{2},\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{3}{2}\right\rangle + \frac{1}{2}\left|\frac{3}{2},\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},-\frac{1}{2}\right\rangle - \frac{1}{2}\left|\frac{3}{2},-\frac{1}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{1}{2}\right\rangle - \frac{1}{2}\left|\frac{3}{2},-\frac{3}{2}\right\rangle\otimes\left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|1,0\rangle = \frac{3}{2\sqrt{5}}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{2\sqrt{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle - \frac{1}{2\sqrt{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle + \frac{3}{2\sqrt{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|0,0\rangle = \frac{1}{2}\left|\frac{3}{2},\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{2}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{1}{2}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle - \frac{1}{2}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{3}{2}\right\rangle$$

$$|3,-1\rangle = \frac{1}{\sqrt{5}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle + \sqrt{\frac{3}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle + \frac{1}{\sqrt{5}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$|2,-1\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$|1,-1\rangle = \sqrt{\frac{3}{10}}\left|\frac{3}{2},\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle - \sqrt{\frac{2}{5}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{3}{10}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},\frac{1}{2}\right\rangle$$

$$|3,-2\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle + \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

$$|2,-2\rangle = \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle - \frac{1}{\sqrt{2}}\left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{1}{2}\right\rangle$$

$$|3,-3\rangle = \left|\frac{3}{2},-\frac{3}{2}\right\rangle \otimes \left|\frac{3}{2},-\frac{3}{2}\right\rangle$$

3.10. $j_1 = 2$ y $j_2 = 2$

$$|4,4\rangle = |2,2\rangle \otimes |2,2\rangle$$

$$|4,3\rangle = \frac{1}{\sqrt{2}}|2,2\rangle \otimes |2,1\rangle + \frac{1}{\sqrt{2}}|2,1\rangle \otimes |2,2\rangle$$

$$|3,3\rangle = \frac{1}{\sqrt{2}}|2,2\rangle \otimes |2,1\rangle - \frac{1}{\sqrt{2}}|2,1\rangle \otimes |2,2\rangle$$

$$|4,2\rangle = \sqrt{\frac{3}{14}}|2,2\rangle \otimes |2,0\rangle + \frac{2}{\sqrt{7}}|2,1\rangle \otimes |2,1\rangle + \sqrt{\frac{3}{14}}|2,0\rangle \otimes |2,2\rangle$$

$$|3,2\rangle = \frac{1}{\sqrt{2}}|2,2\rangle \otimes |2,0\rangle - \frac{1}{\sqrt{2}}|2,0\rangle \otimes |2,2\rangle$$

$$|2,2\rangle = \sqrt{\frac{2}{7}}|2,2\rangle \otimes |2,0\rangle - \sqrt{\frac{3}{7}}|2,1\rangle \otimes |2,1\rangle + \sqrt{\frac{2}{7}}|2,0\rangle \otimes |2,2\rangle$$

$$|4,1\rangle = \frac{1}{\sqrt{14}}\,|2,2\rangle \otimes |2,-1\rangle + \sqrt{\frac{3}{7}}\,|2,1\rangle \otimes |2,0\rangle + \sqrt{\frac{3}{7}}\,|2,0\rangle \otimes |2,1\rangle + \frac{1}{\sqrt{14}}\,|2,-1\rangle \otimes |2,2\rangle$$

$$|3,1\rangle = \sqrt{\frac{3}{10}}\,|2,2\rangle \otimes |2,-1\rangle + \frac{1}{\sqrt{5}}\,|2,1\rangle \otimes |2,0\rangle - \frac{1}{\sqrt{5}}\,|2,0\rangle \otimes |2,1\rangle - \sqrt{\frac{3}{10}}\,|2,-1\rangle \otimes |2,2\rangle$$

$$|2,1\rangle = \sqrt{\frac{3}{7}}\,|2,2\rangle \otimes |2,-1\rangle - \frac{1}{\sqrt{14}}\,|2,1\rangle \otimes |2,0\rangle - \frac{1}{\sqrt{14}}\,|2,0\rangle \otimes |2,1\rangle + \sqrt{\frac{3}{7}}\,|2,-1\rangle \otimes |2,2\rangle$$

$$|1,1\rangle = \frac{1}{\sqrt{5}}\,|2,2\rangle \otimes |2,-1\rangle - \sqrt{\frac{3}{10}}\,|2,1\rangle \otimes |2,0\rangle + \sqrt{\frac{3}{10}}\,|2,0\rangle \otimes |2,1\rangle - \frac{1}{\sqrt{5}}\,|2,-1\rangle \otimes |2,2\rangle$$

$$|4,0\rangle = \frac{1}{\sqrt{70}}\,|2,2\rangle \otimes |2,-2\rangle + 2\sqrt{\frac{2}{35}}\,|2,1\rangle \otimes |2,-1\rangle + 3\sqrt{\frac{2}{35}}\,|2,0\rangle \otimes |2,0\rangle + 2\sqrt{\frac{2}{35}}\,|2,-1\rangle \otimes |2,1\rangle + \frac{1}{\sqrt{70}}\,|2,-2\rangle \otimes |2,2\rangle$$

$$|3,0\rangle = \frac{1}{\sqrt{10}}\,|2,2\rangle \otimes |2,-2\rangle + \sqrt{\frac{2}{5}}\,|2,1\rangle \otimes |2,-1\rangle - \sqrt{\frac{2}{5}}\,|2,-1\rangle \otimes |2,1\rangle - \frac{1}{\sqrt{10}}\,|2,-2\rangle \otimes |2,2\rangle$$

$$|2,0\rangle = \sqrt{\frac{2}{7}}\,|2,2\rangle \otimes |2,-2\rangle + \frac{1}{\sqrt{14}}\,|2,1\rangle \otimes |2,-1\rangle - \sqrt{\frac{2}{7}}\,|2,0\rangle \otimes |2,0\rangle + \frac{1}{\sqrt{14}}\,|2,-1\rangle \otimes |2,1\rangle + \sqrt{\frac{2}{7}}\,|2,-2\rangle \otimes |2,2\rangle$$

$$|1,0\rangle = \sqrt{\frac{2}{5}}\,|2,2\rangle \otimes |2,-2\rangle - \frac{1}{\sqrt{10}}\,|2,1\rangle \otimes |2,-1\rangle + \frac{1}{\sqrt{10}}\,|2,-1\rangle \otimes |2,1\rangle - \sqrt{\frac{2}{5}}\,|2,-2\rangle \otimes |2,2\rangle$$

$$|0,0\rangle = \frac{1}{\sqrt{5}}\,|2,2\rangle \otimes |2,-2\rangle - \frac{1}{\sqrt{5}}\,|2,1\rangle \otimes |2,-1\rangle + \frac{1}{\sqrt{5}}\,|2,0\rangle \otimes |2,0\rangle - \frac{1}{\sqrt{5}}\,|2,-1\rangle \otimes |2,1\rangle + \frac{1}{\sqrt{5}}\,|2,-2\rangle \otimes |2,2\rangle$$

$$|4,-1\rangle = \frac{1}{\sqrt{14}}\,|2,1\rangle \otimes |2,-2\rangle + \sqrt{\frac{3}{7}}\,|2,0\rangle \otimes |2,-1\rangle + \sqrt{\frac{3}{7}}\,|2,-1\rangle \otimes |2,0\rangle + \frac{1}{\sqrt{14}}\,|2,-2\rangle \otimes |2,1\rangle$$

$$|3,-1\rangle = \sqrt{\frac{3}{10}}\,|2,1\rangle\otimes|2,-2\rangle + \frac{1}{\sqrt{5}}\,|2,0\rangle\otimes|2,-1\rangle - \frac{1}{\sqrt{5}}\,|2,-1\rangle\otimes|2,0\rangle - \sqrt{\frac{3}{10}}\,|2,-2\rangle\otimes|2,1\rangle$$

$$|2,-1\rangle = \sqrt{\frac{3}{7}}\,|2,1\rangle\otimes|2,-2\rangle - \frac{1}{\sqrt{14}}\,|2,0\rangle\otimes|2,-1\rangle - \frac{1}{\sqrt{14}}\,|2,-1\rangle\otimes|2,0\rangle + \sqrt{\frac{3}{7}}\,|2,-2\rangle\otimes|2,1\rangle$$

$$|1,-1\rangle = \frac{1}{\sqrt{5}}\,|2,1\rangle\otimes|2,-2\rangle - \sqrt{\frac{3}{10}}\,|2,0\rangle\otimes|2,-1\rangle + \sqrt{\frac{3}{10}}\,|2,-1\rangle\otimes|2,0\rangle - \frac{1}{\sqrt{5}}\,|2,-2\rangle\otimes|2,1\rangle$$

$$|4,-2\rangle = \sqrt{\frac{3}{14}}\,|2,0\rangle\otimes|2,-2\rangle + \frac{2}{\sqrt{7}}\,|2,-1\rangle\otimes|2,-1\rangle + \sqrt{\frac{3}{14}}\,|2,-2\rangle\otimes|2,0\rangle$$

$$|3,-2\rangle = \frac{1}{\sqrt{2}}\,|2,0\rangle\otimes|2,-2\rangle - \frac{1}{\sqrt{2}}\,|2,-2\rangle\otimes|2,0\rangle$$

$$|2,-2\rangle = \sqrt{\frac{2}{7}}\,|2,0\rangle\otimes|2,-2\rangle - \sqrt{\frac{3}{7}}\,|2,-1\rangle\otimes|2,-1\rangle + \sqrt{\frac{2}{7}}\,|2,-2\rangle\otimes|2,0\rangle$$

$$|4,-3\rangle = \frac{1}{\sqrt{2}}\,|2,-1\rangle\otimes|2,-2\rangle + \frac{1}{\sqrt{2}}\,|2,-2\rangle\otimes|2,-1\rangle$$

$$|3,-3\rangle = \frac{1}{\sqrt{2}}\,|2,-1\rangle\otimes|2,-2\rangle - \frac{1}{\sqrt{2}}\,|2,-2\rangle\otimes|2,-1\rangle$$

$$|4,-4\rangle = |2,-2\rangle\otimes|2,-2\rangle$$